

Tidyverse cont'd

Data Visualization for Social Good

CorrelAid Switzerland



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The R *tidyverse* can do a lot more

1. *tidyverse* core
2. *tidyverse* extended



tidyverse core

1 The tidyverse is...

- A collection of user-friendly **packages** for analyzing **tidy data**
- An **ecosystem** for analytics and data science with common design principles
- A **dialect** of the R language



tidyverse extended

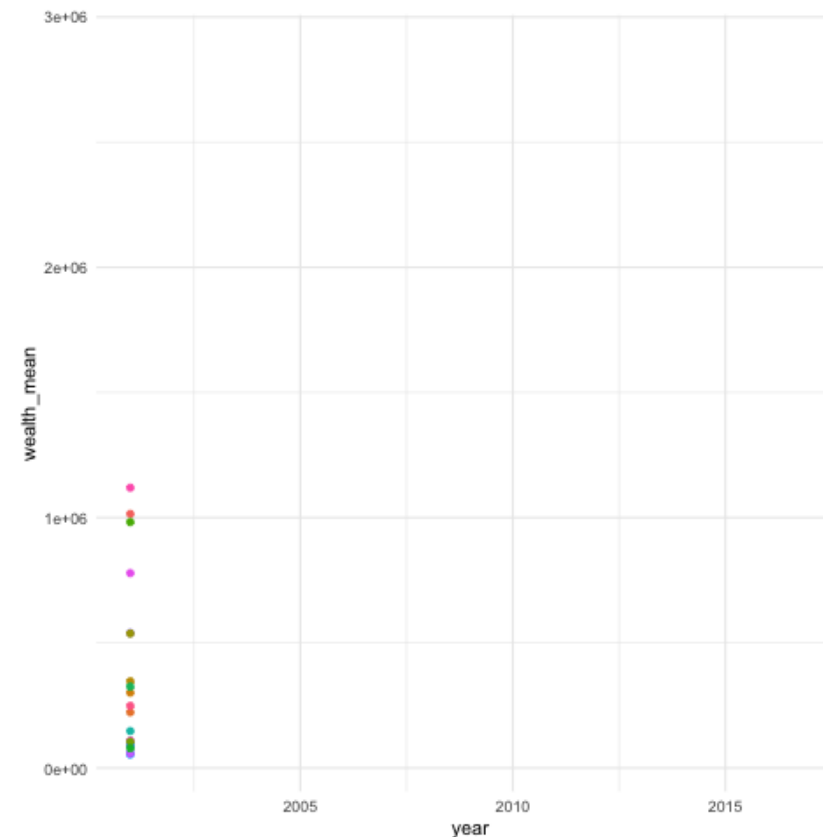
- 1 The tidyverse is...an **ecosystem** for analytics and data science with common design principles





gganimate for animations

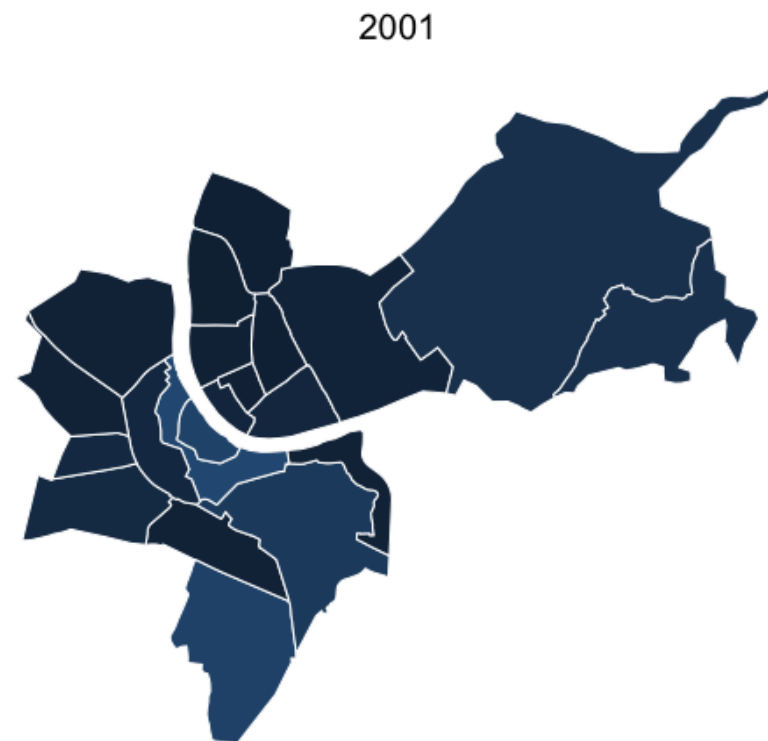
```
basel %>%  
  ggplot(aes(x = year, col = quarter)) +  
  geom_point(aes(y = wealth_mean)) +  
  geom_line(aes(y = wealth_mean)) +  
  guides(col = FALSE) +  
  theme_minimal() +  
  
  # add animation  
  transition_reveal(year)
```





gganimate for animations

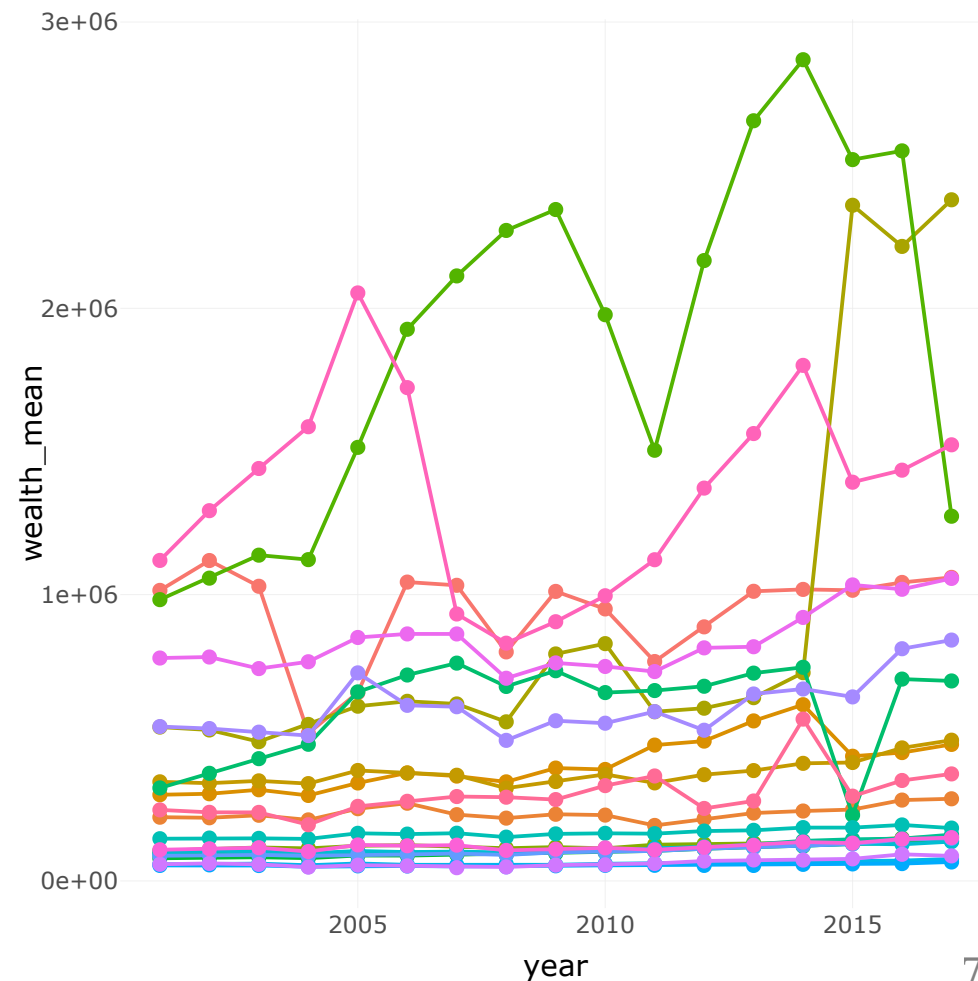
```
plot = basel_map %>%  
  
  # original code  
  left_join(basel,  
            by = c('TYPE' = 'quarter')) %>%  
  ggplot(aes(fill = wealth_mean)) +  
  geom_sf(col = 'white') +  
  guides(fill = FALSE) +  
  theme_void() +  
  
  # add animation  
  ggtitle("{closest_state}") +  
  theme(plot.title = element_text(  
    hjust = 0.5, size=20)) +  
  transition_states(year)
```



plotly for interactive graphics

```
# original plot
plot <- basel_map %>%
  left_join(basel,
            by = c('TYPE' = 'quarter')) %>%
  ggplot(aes(x = year, col = TYPE)) +
  geom_point(aes(y = wealth_mean)) +
  geom_line(aes(y = wealth_mean)) +
  guides(col = FALSE) +
  theme_minimal()

# plotlify
ggplotly(plot)
```





RMarkdown for reports

```
---  
title: "Basel"  
output: pdf_document  
---
```

```
```{r setup, include=FALSE}  
knitr::opts_chunk$set(echo = FALSE)

Read data
library(tidyverse)
library(sf)
basel_map = sf::read_sf('1_Data/map')
basel <- read_csv('1_Data/taxation.csv')
```
```

Wealth map

The average wealth is $\text{mean}(\text{basel}\$income_mean)$.

This is a map of wealth in 2017.

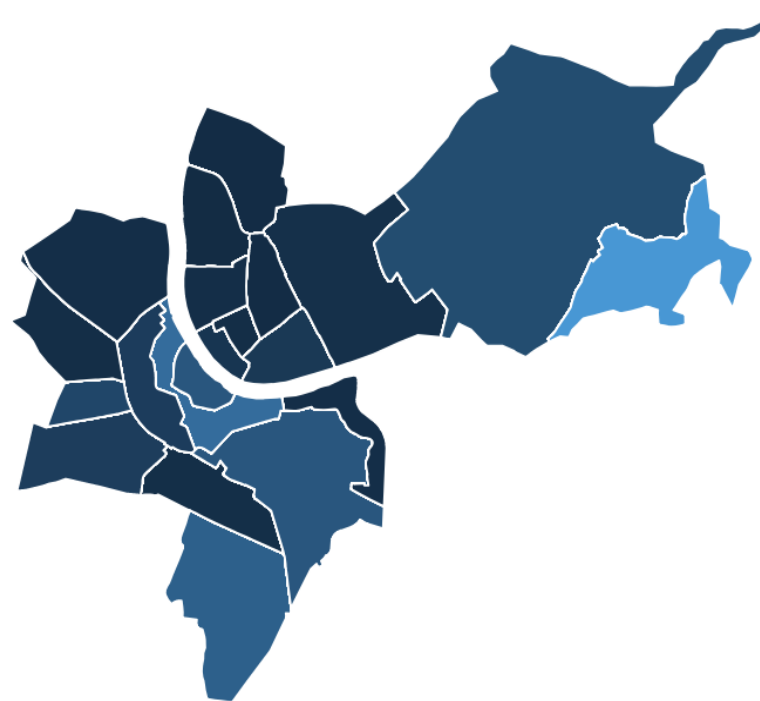
```
```{r, message=FALSE, fig.height=4, fig.width=4}  
Wealth map
basel_map %>%
 left_join(basel, by = c('TYPE' = 'quarter')) %>%
 ggplot() +
 geom_sf(col = 'white',
 mapping = aes(fill = wealth_mean)) +
 guides(fill = FALSE) +
 theme_void()
```
```

Basel

Wealth map

The average wealth is 6.6774073×10^4 .

This is a map of wealth in 2017.




```
ui <- fluidPage(  
  tabsetPanel(  
    tabPanel(  
      selectInput(  
        "var", "Variable",  
        choices = list("wealth_mean",  
                        "wealth_median",  
                        "wealth_gini"))  
      mainPanel(plotOutput("basel"))  
    )  
  )  
  
server <- function(input, output) {  
  output$basel <- renderPlot({  
    ggplot(map) +  
    geom_sf(  
      col = 'white',  
      mapping = aes_string(  
        fill = input$var)) +  
    guides(fill = FALSE) +  
    theme_void()})  
}
```



rvest for web scraping

```
# Tabelle laden von Wikipedia (Pakete nicht vergessen)
read_html("https://en.wikipedia.org/wiki/R_(programming_language)") %>%
  html_node(xpath = '//*[@id="mw-content-text"]/div[1]/table[2]') %>%
  html_table() %>% as_tibble()
```

```
## # A tibble: 21 x 3
```

```
##   Release Date      Description
```

```
##   <chr>      <chr>      <chr>
```

```
## 1 0.16      ""          "This is the last alpha version developed primarily by Ihaka and G...
```

```
## 2 0.49      "1997-04-2... "This is the oldest source release which is currently available on...
```

```
## 3 0.60      "1997-12-0... "R becomes an official part of the GNU Project. The code is hosted...
```

```
## 4 0.65.1    "1999-10-0... "First versions of update.packages and install.packages functions ...
```

```
## 5 1.0       "2000-02-2... "Considered by its developers stable enough for production use.[55...
```

```
## 6 1.4       "2001-12-1... "S4 methods are introduced and the first version for Mac OS X is m...
```

```
## 7 1.8       "2003-10-0... "Introduced a flexible condition handling mechanism for signalling...
```

```
## 8 2.0       "2004-10-0... "Introduced lazy loading, which enables fast loading of data with ...
```

```
## 9 2.1       "2005-04-1... "Support for UTF-8 encoding, and the beginnings of internationaliz...
```

```
## 10 2.6.2    "2008-02-0... "Last version to support Windows 95, 98, Me and NT 4.0[56]"
```

```
## # ... with 11 more rows
```



```
# Wörter zählen
counts <-
  read_html(".../R_(programming_language)") %>%
  html_text() %>%
  tibble() %>%
  unnest_tokens(w, ".") %>%
  filter(!str_detect(w, '[:digit:]')) %>%
  anti_join(stop_words) %>%
  count(word, sort = TRUE) %>%
  top_n(100)

# Wordwolke
wordcloud(counts$word,
           counts$n)
```





tidymodels for machine learning

- 1 Framework for machine learning.
- 2 Includes
 - All sorts of models, including wrapper to **keras**, for deep learning.
 - Model tuning
 - Model evaluation
 - etc.





dbplyr

for wrangling remote databases

- 1 R tidyverse is able to interface with most database types: MySQL, MariaDB, BigQuery, Redshift, etc. See db.rstudio.com.
- 2 dplyr will be **translated in the background** to the appropriate query language

```
# Verbinde mit MySQL Datenbank
con <- dbConnect(MySQL(),
  user='studiech_rbootca',
  password='Du*5hA+7NU:8T',
  dbname='studiech_rbootcamp',
  host='studie.ch',
  port = 3306)
```

```
# Extrahiere Tabelle Customers
basel <- tbl(con, "basel")
basel
```

```
## # Source:   table<basel> [?? x 20]
## # Database: mysql 5.7.26-log-cll-lve
## #   [studiech_rbootca@studie.ch:/studiech_rbootc
##       id geschlecht alter_jahre groesse
##   <dbl> <chr>          <dbl>    <dbl>
## 1      1 f              87      165
## 2      2 m              54      175.
## 3      3 f              34      147.
## 4      4 m              31      166.
## 5      5 m              24      180.
## # ... with more rows, and 16 more
## #   variables
```

Books

An incomplete list of good books loosely sorted by required amount of R experience.



Thank you!